

Energy Consumption of Modern CPUs and GPUs: A Study with Production Applications

Bachelor Thesis First Talk

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Agenda

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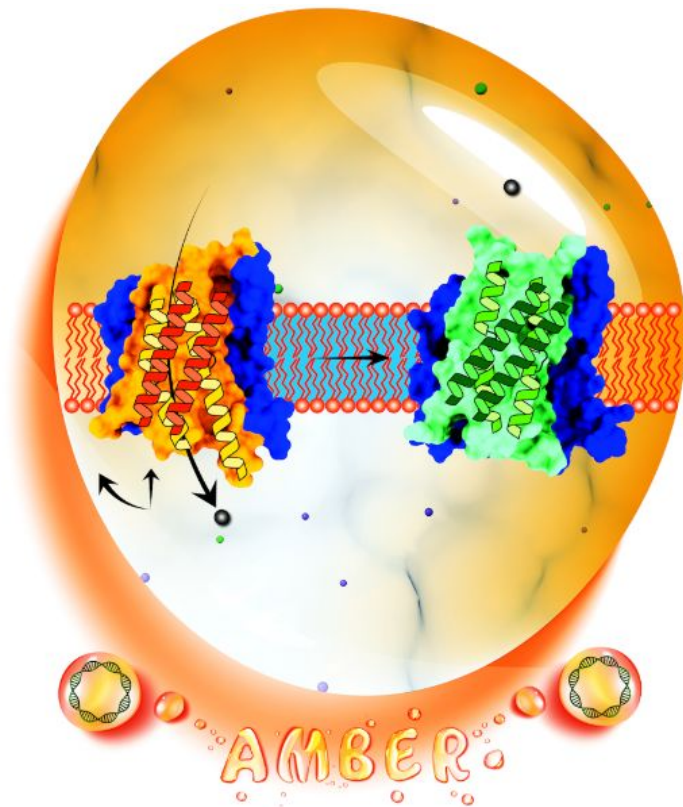
Objective of the thesis

- For MD workloads (GROMACS, AMBER, LAMMPS), how do modern CPUs and GPUs compare in time-to-solution and energy-to-solution under realistic HPC configurations?
- How do GPU frequency and GPU power-cap settings affect
 - energy-to-solution
 - time-to-solution
 - and how can we optimize that
- How do CPU frequency and MPI rank count affect
 - energy-to-solution
 - time-to-solution
 - scaling limitations

Application

AMBER

- Biomolecular molecular dynamics simulation Suite
- Widely used by many
- Heterogeneous support for both CPU and GPU execution
- pmemd.cuda
 - GPU-accelerated pmemd
- sander.MPI
 - General MD engine
 - MPI-parallel CPU implementation
- pmemd.MPI
 - Optimized production MD engine
 - Typically higher performance



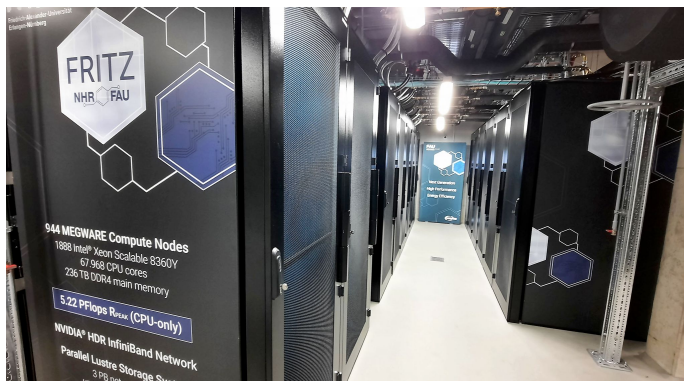
Experimental Setup



Hardware

Fritz

- Intel Xeon Platinum 8360Y
- @2.4 GHz
- IceLake architecture (ICX)
- 2 processors per node (36 + 36), 4 ccNUMA domains per node
- 256GB memory (DDR4), 409.6 GB/s
- 54 MB L3 cache



Alex

- AMD EPYC 7713 @2.0 GHz
- Zen 3 Milan architecture
- 2 processors per node 64 x 2,
- **8x Nvidia A100 (40 GB HBM2)**
 - 1 TB RAM
- 8x Nvidia A100 (80 GB HBM2)
 - 2 TB RAM
- **8x Nvidia A40 (48 GB DDR6)**
 - 512 GB RAM



Software

Execution environment: Slurm (sbatch + srun)

GPU benchmarks

Module

- amber/24p02-at24p05-gnu-cuda12.4

Binary

- pmemd.CUDA

GPU control/telemetry

- nvidia smi

CPU benchmarks

Module

- amber/20p12-at21p11-impi-intel

Binaries

- sander.MPI
- pmemd.MPI

CPU Control/telemetry

- likwid/5.4.1

Workloads

Input GPU	Number of Atoms
JAC (NPT/NVE)	23558
Factorix (NPT/NVE)	90906
STMV (NPT/NVE)	1067095
Cellulose (NPT/NVE)	408609
TRPCage	304
myoglobin	2492
nucleosome	25095

Input CPU	Number of Atoms
Factorix	90906
Dhfr	22930
Jac	23558
Trx	14093

Methodology

Alex

- GPU frequency sweep:
 - A100: 240–1410 MHz (step 45 MHz)
 - A40: 240–1780 MHz (step 60 MHz)
- GPU power-cap sweep:
 - A100: 100–400 W (step 15 W)
 - A40: 100–300 W (step 15 W)

Measurement

- `--loop-ms=100`
- `--query-gpu=timestamp,power.draw,utilization.gpu`

Adjusting the frequency/powercap

- `--applications-clocks=${MemFreq}-${ClockFreq}`
- `--power-limit=${Limit}`

Fritz

For scaling/energy/performance

- sweep over 1 ccNUMA domain (2-18 MPI processes)
- base 2.4GHz, unless stated otherwise
- 250 W TDP, no power cap applied

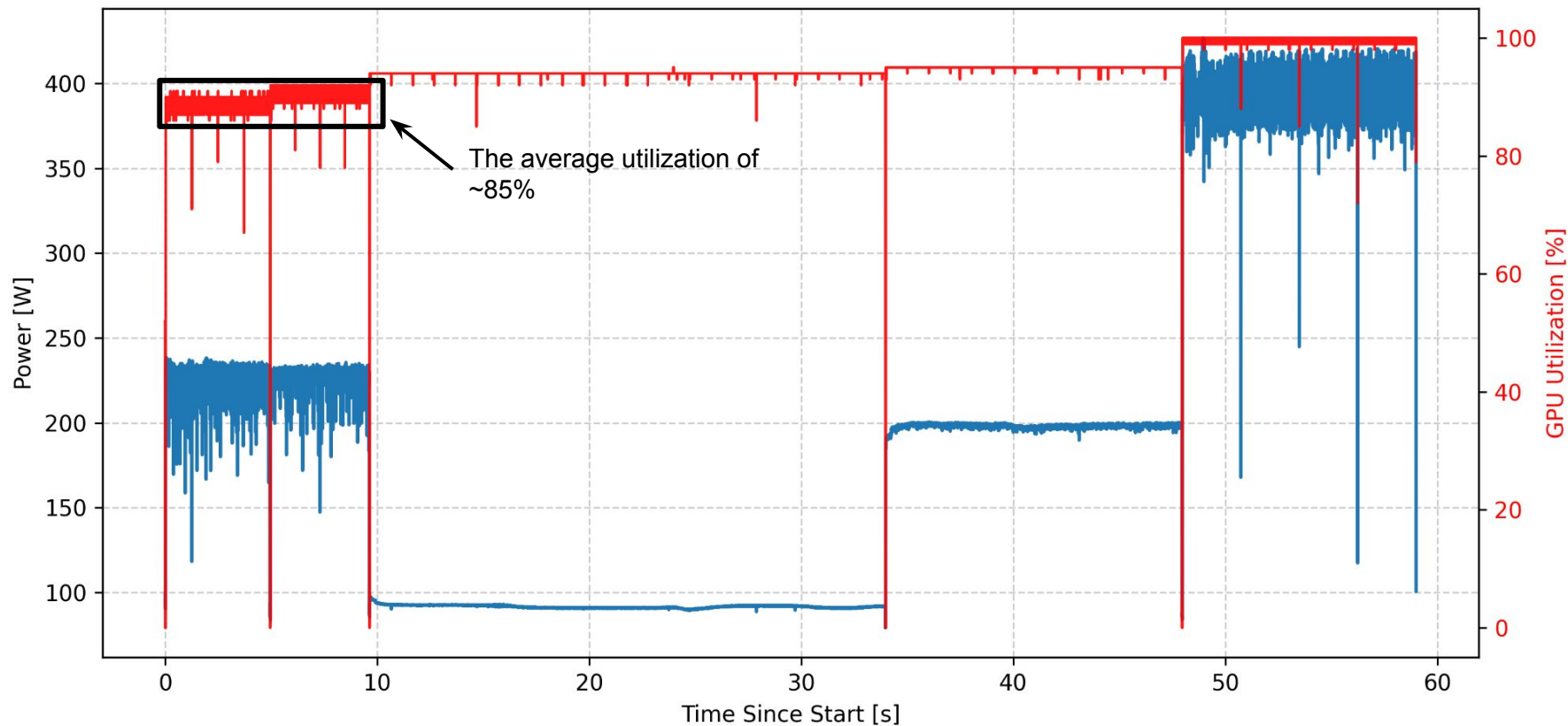
Measurement

- `-g Energy`
- `-g MEM_DP`

Adjusting Parameters

- `--mpiopts "--cpu-freq=${freq}-${freq}:performance \
--cpu-bind=map_cpu:${cpumap} \
--distribution=block:block" \
-np ${tmpi}`

GPU Power Draw and Utilization over Time



Methodology

Alex

- GPU frequency sweep:
 - A100: 240–1410 MHz (step 60 MHz)
 - A40: 240–1780 MHz (step 60 MHz)
- GPU power-cap sweep:
 - A100: 100–400 W (step 15 W)
 - A40: 100–300 W (step 15 W)

Measurement

- --loop-ms=100
- --query-gpu=timestamp,power.draw,utilization.gpu

Adjusting the frequency/powercap

- --applications-clocks=\${MemFreq}-\${ClockFreq}
- --power-limit=LIMIT

Fritz

For scaling/energy/performance

- sweep over 1 ccNUMA domain (2-18 MPI processes)
- base 2.4GHz, unless stated otherwise
- 250 W TDP, no power cap applied

Measurement

- -g Energy
- -g MEM_DP

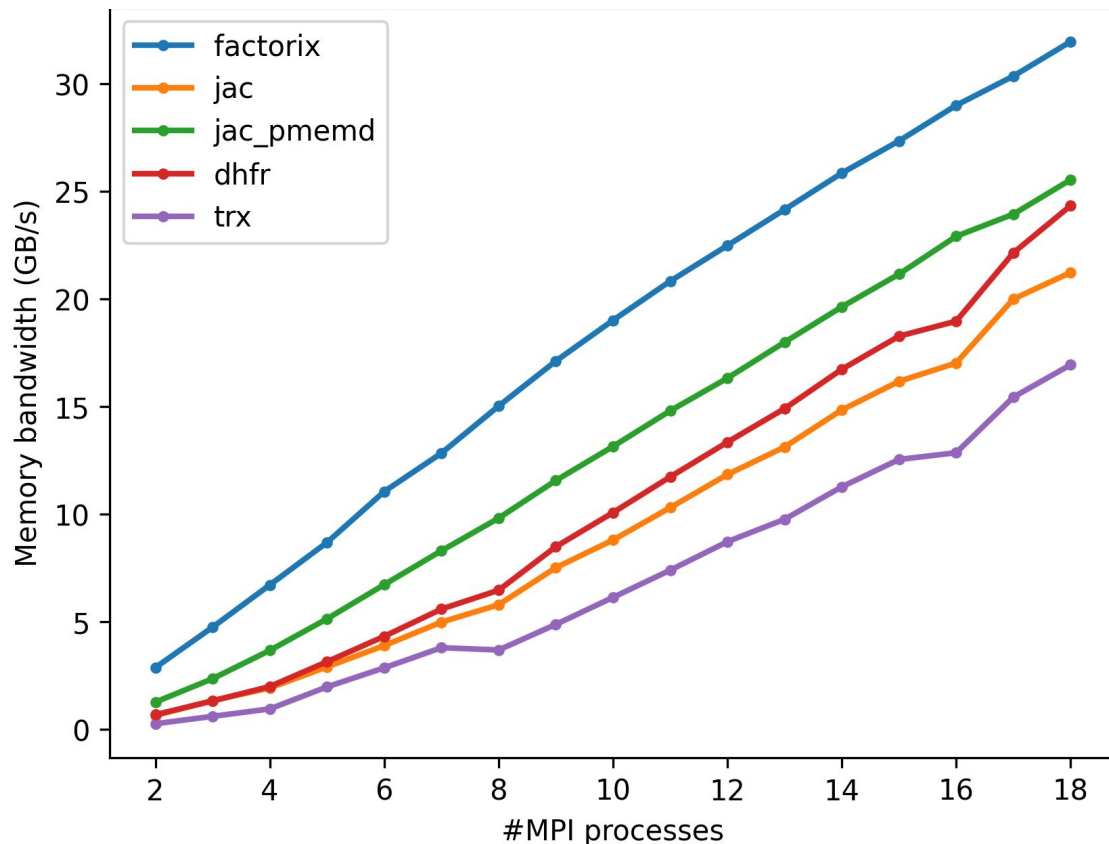
Adjusting Parameters

- --mpiopts "--cpu-freq=\${freq}-\${freq}:performance \
--cpu-bind=map_cpu:\${cpumap} \
--distribution=block:block" \
-np \${tmpi}

CPU Results

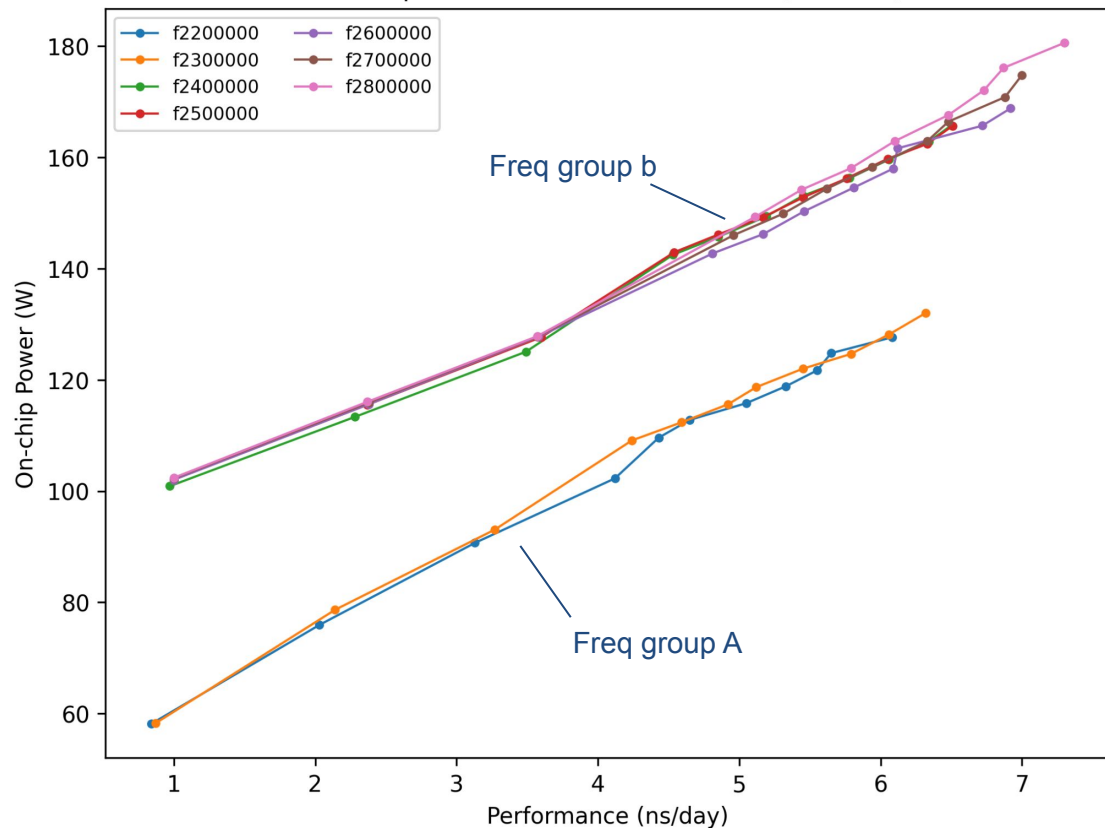


Memory Bandwidth vs. MPI Processes



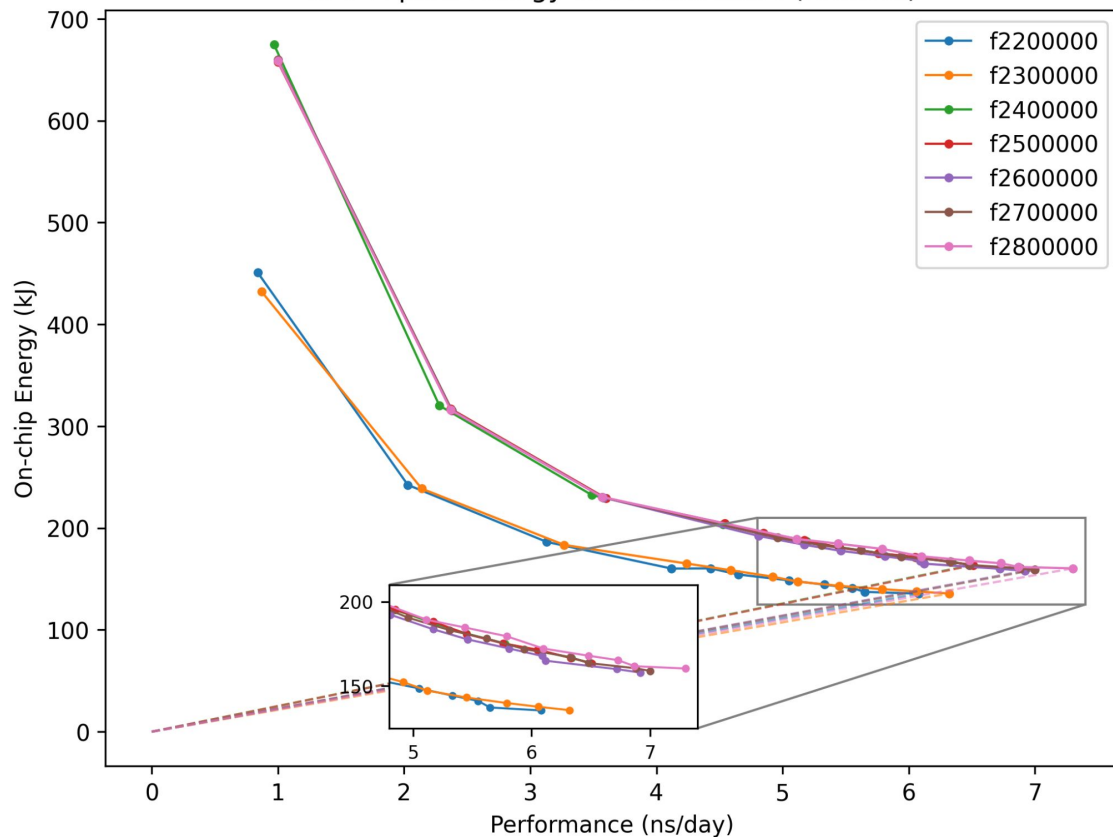
- No clear Memory saturation within 1 ccNuma domain
- Fritz **theoretical memory bandwidth** is 204.8 GB/s per socket.
- Not an memory bound Application

Frequency Impact on Performance-Wattage: Factorix



- Two distinct power regimes between 2.2-2.3GHz and 2.4-2.8GHz
- Both show similar trend in linear power usage further indicating a compute bound application.
- TDP of 250W has not been reached
 - 70% usage of TDP in highest configuration achieved

Frequency Impact on Performance-Energy Tradeoff

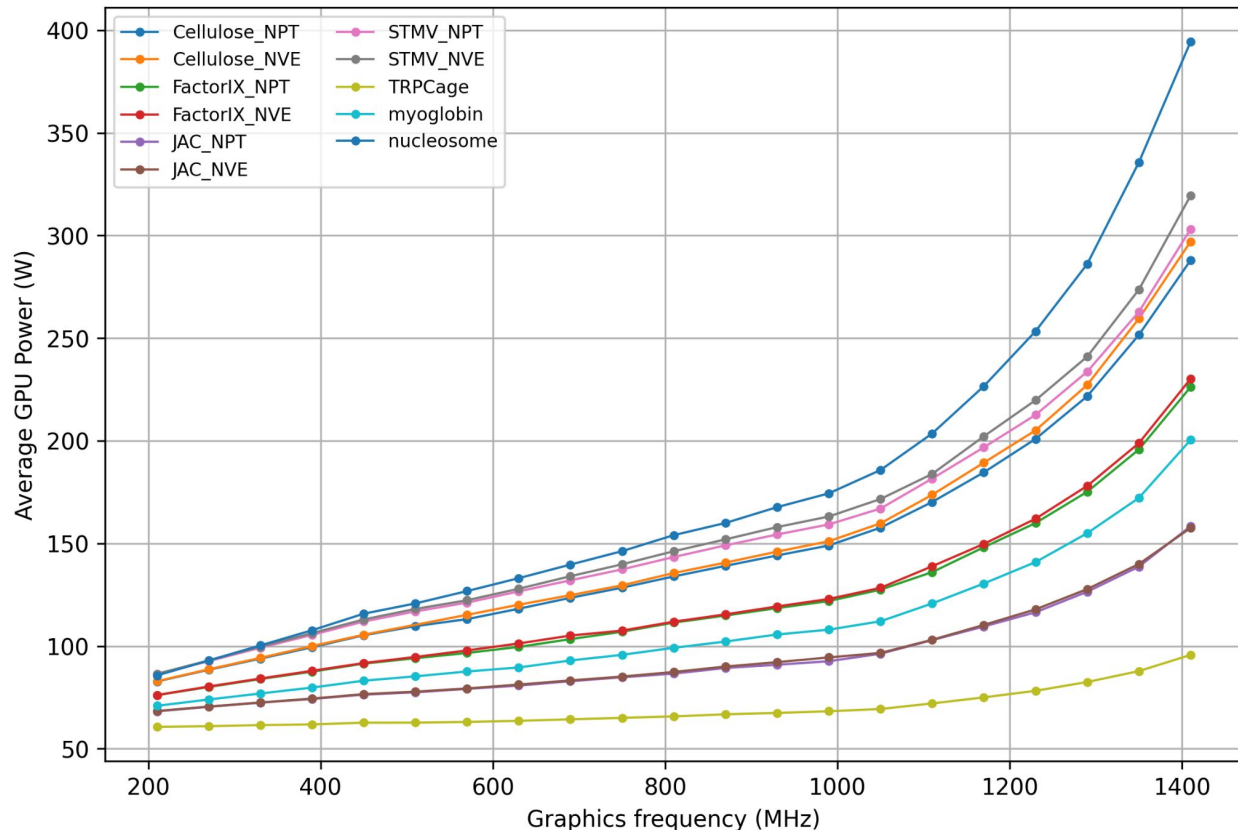


- increasing MPI ranks (within one ccNUMA domain) substantially improves energy-to-solution and performance.
- varying frequency changes efficiency only marginally within a frequency group
- optimal frequency for efficiency is 2.3 GHz

GPUs Result (A100, A40): Frequency Impact



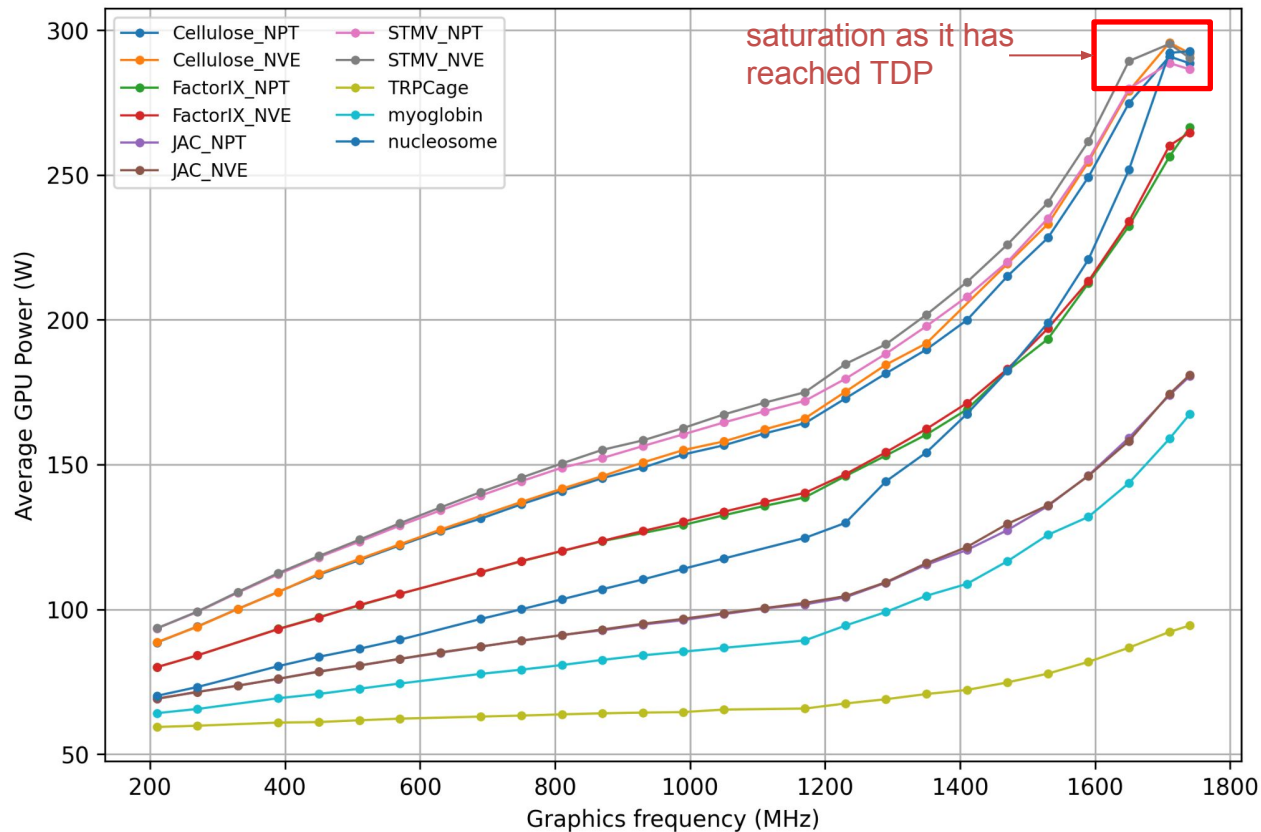
Frequency Impact on Power: A100



1 GHz threshold

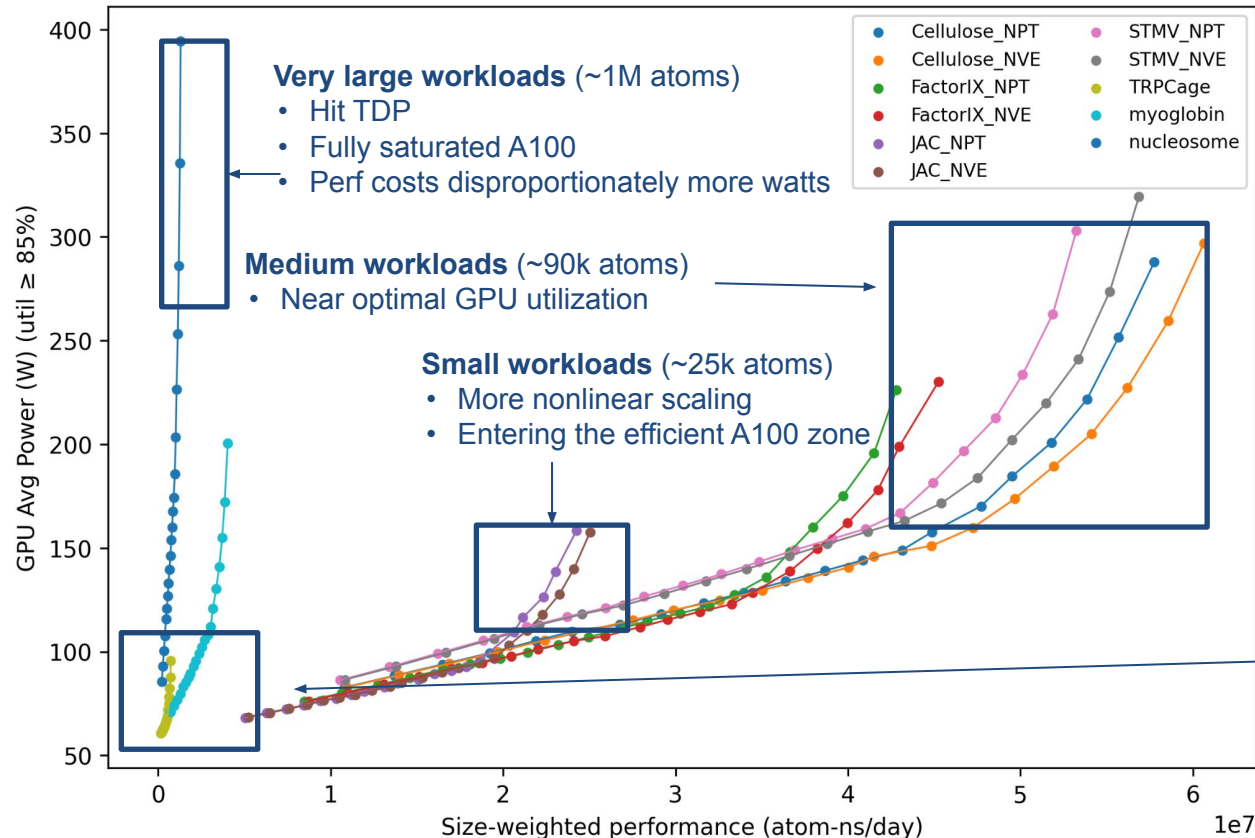
- Below 1 GHz: power rises linearly with graphics frequency
- Above 1 GHz: beyond linear power usage (almost cubic)

Frequency Impact on Power: A40



- Threshold sits at 1.2 GHz

Frequency Impact on Performance-Energy Tradeoff: A100



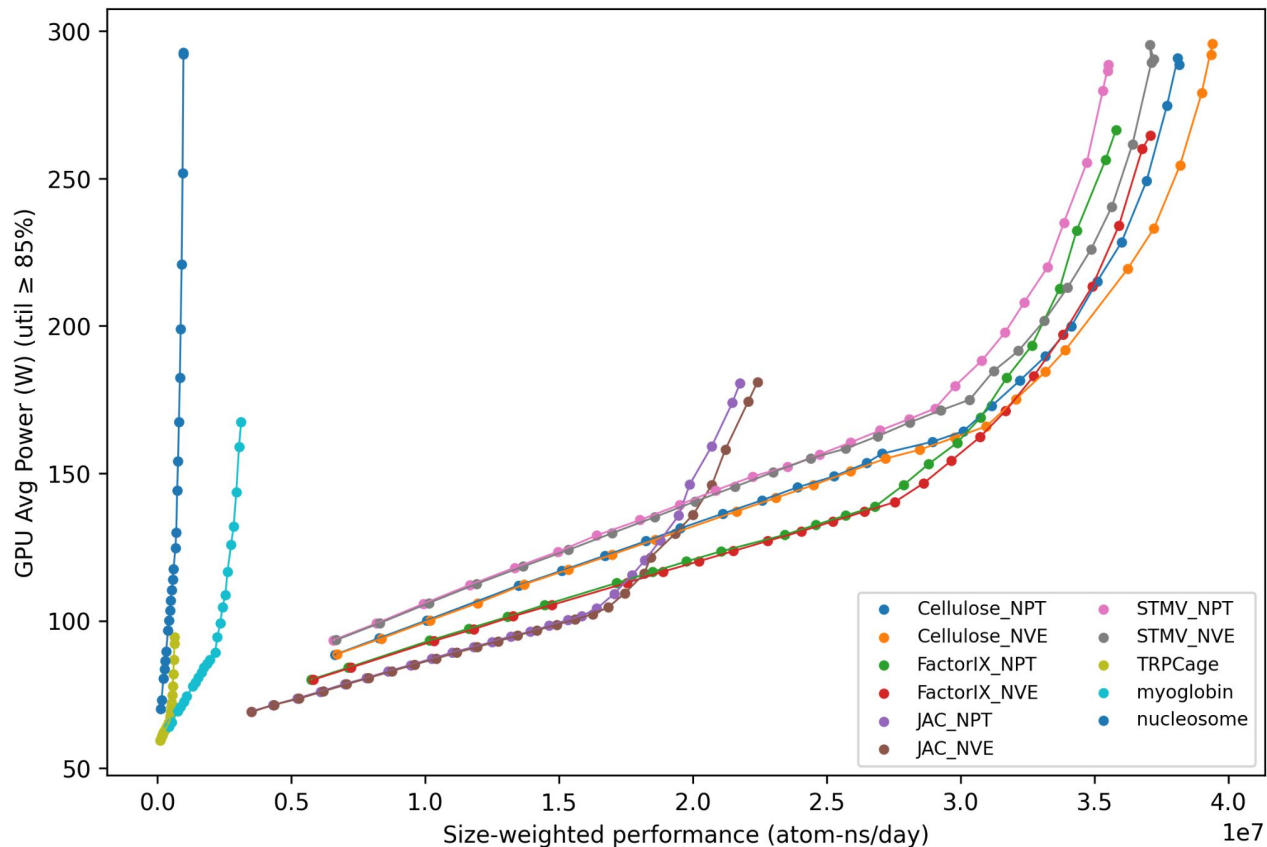
A100 efficiency improves with atom count up to a point:

- <10k atoms → underutilized
- 20k–100k atoms → sweet spot
- 50k – 200k atoms → **Optimal energy efficiency**
- >400k atoms → fully saturated, diminishing returns

Tiny workloads (304 atoms; low power, very high ns/day, power increases slowly)

- A100 is massively underutilized
- Kernel launch overhead dominates
- Memory latency dominates
- Compute units are mostly idle
- A100 is simply too large
- Excellent performance per watt
- Terrible hardware utilization
- Not cost efficient per GPU hour

Frequency Impact on Performance-Energy Tradeoff: A40



Because A40 has

- Fewer SMs
- Lower bandwidth
- Lower peak throughput
- Lower TDP

It saturates earlier and its entire efficiency curve is shifted to the left

A100 efficient region:

~50k – 200k atoms

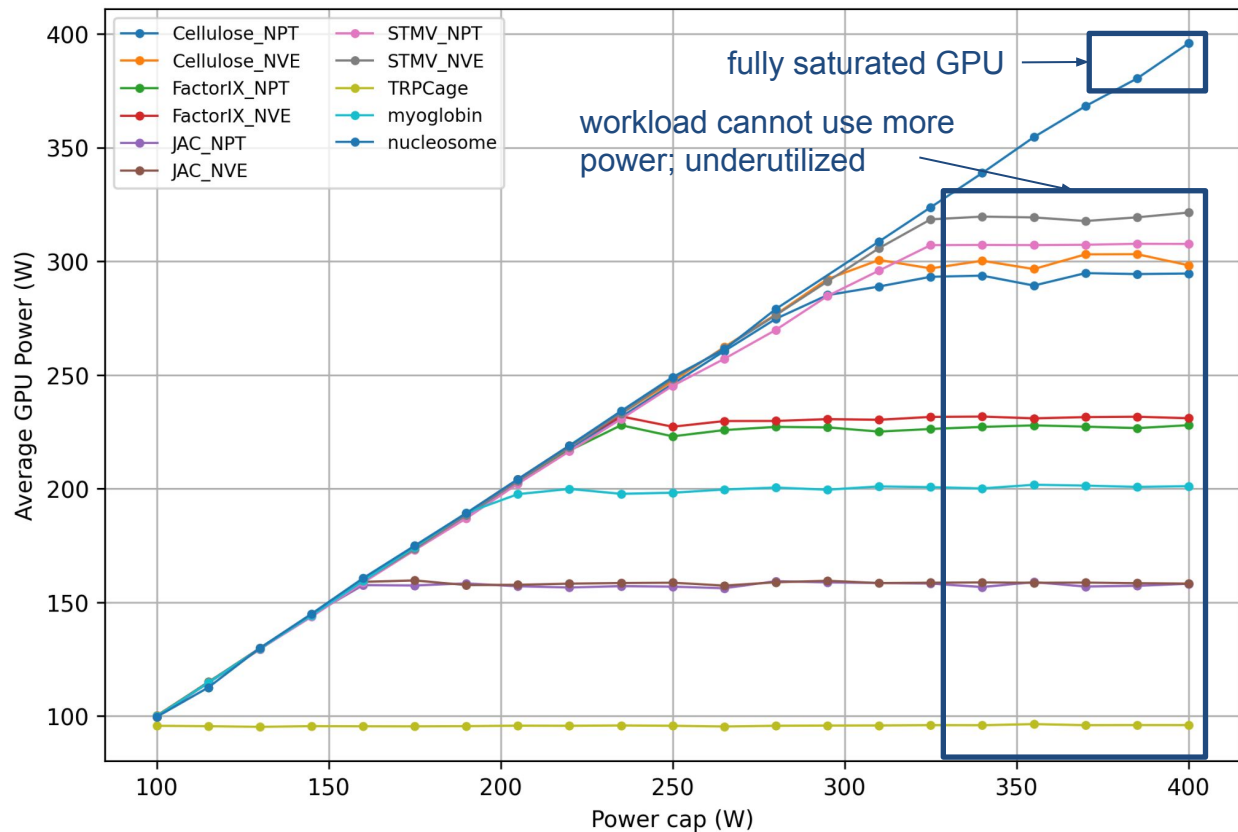
A40 efficient region:

~20k – 80k atoms

GPUs Result (A100, A40): Powercap Impact



Powercap Impact on Power: A100

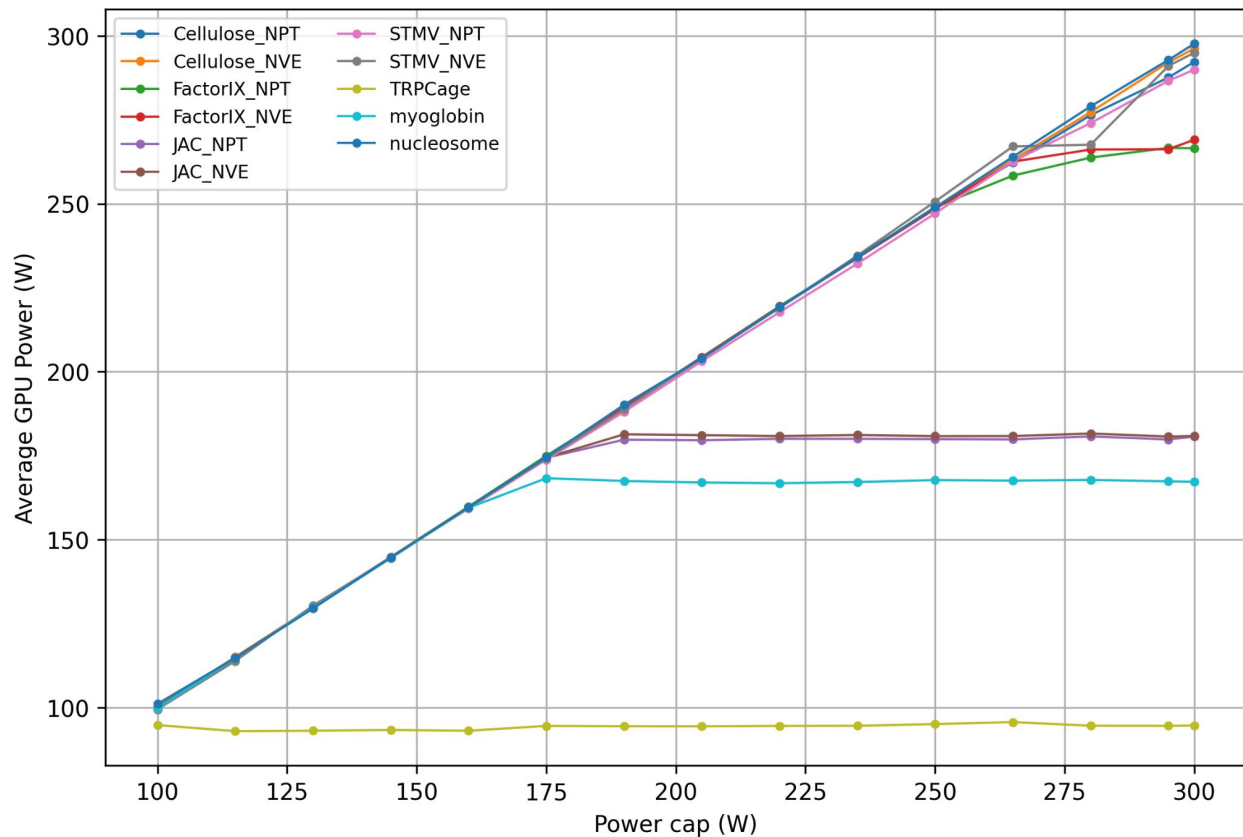


Tiny workloads: Power cap does nothing

Set power cap to:

- * 200–250 W for <100k atoms
- * 275–325 W for 100k–300k atoms
- * run at full 400 W For >400k atoms

Powercap Impact on Power: A40



A100 optimal region

~100k – 400k atoms

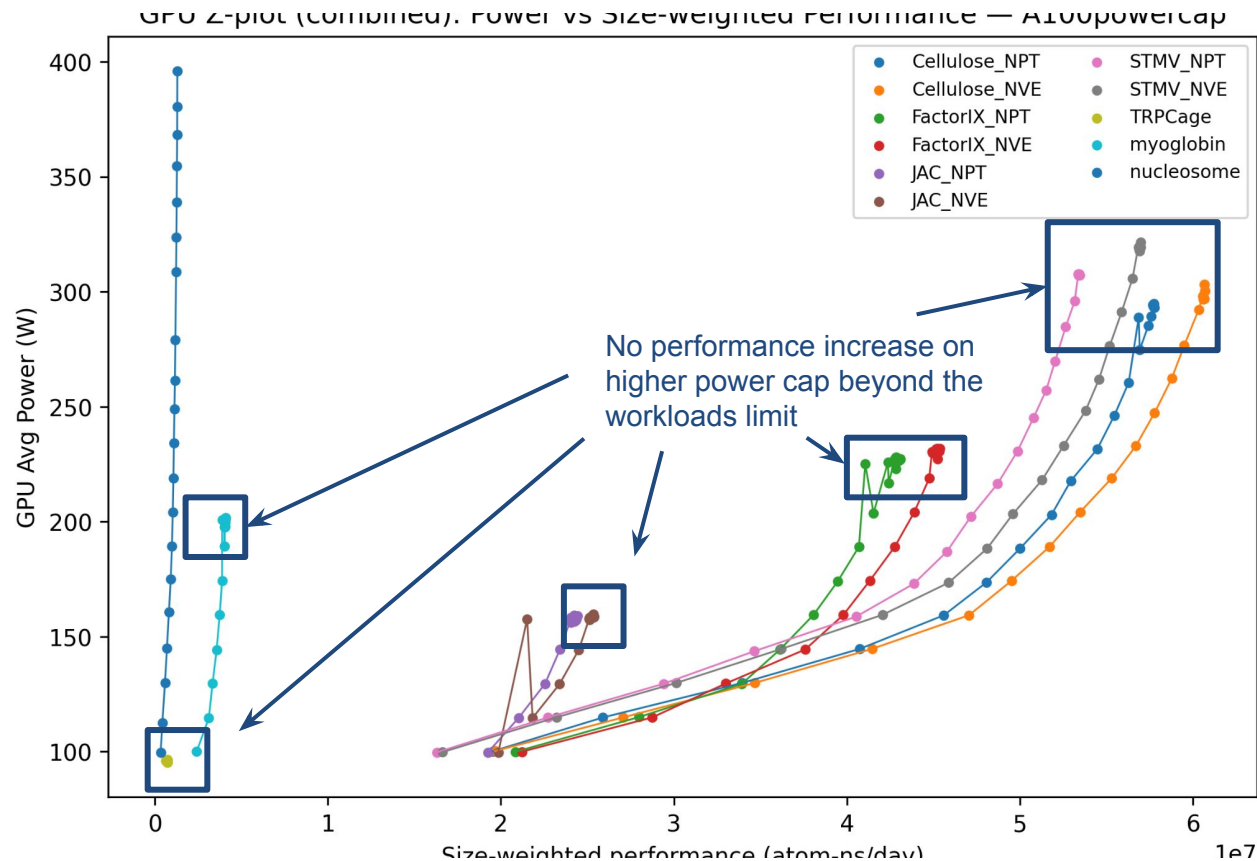
Power cap sweet spot: ~275–325 W

A40 optimal region

~20k – 100k atoms

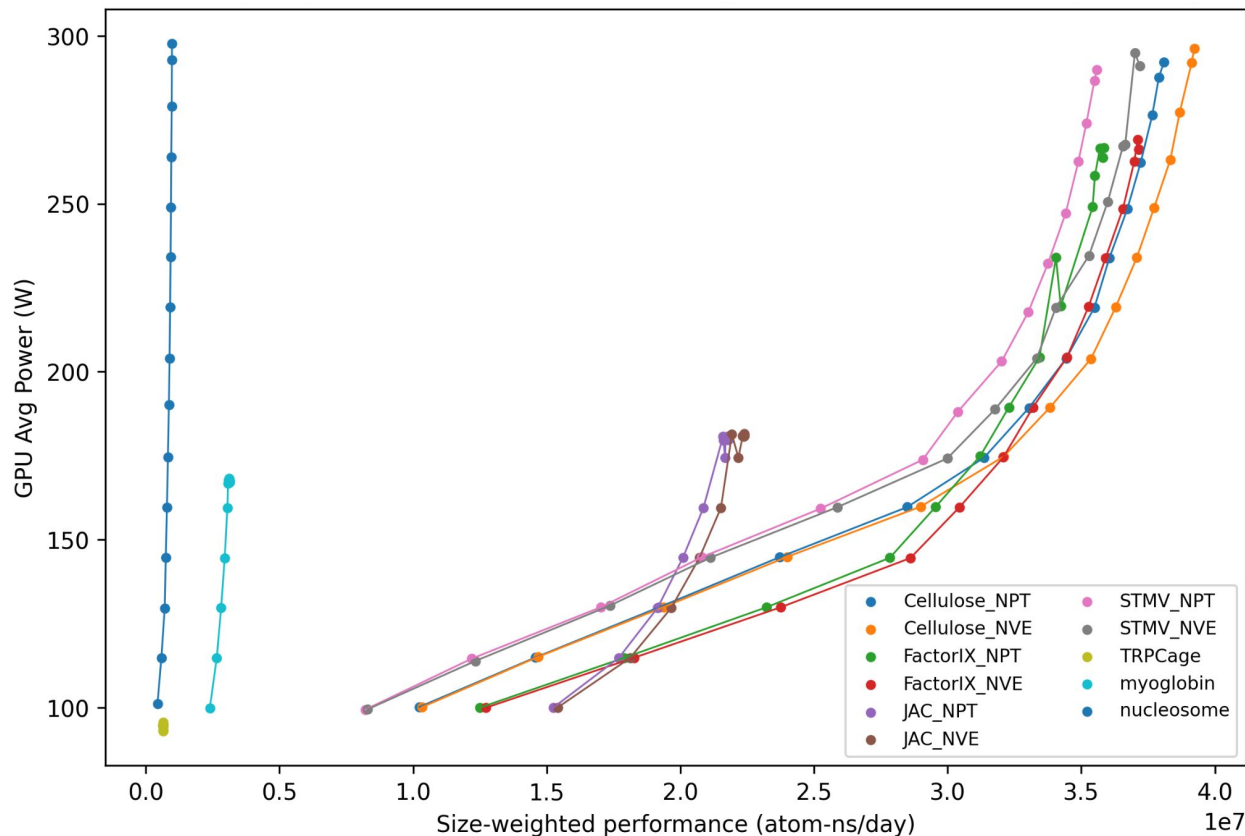
Power cap sweet spot: ~240–270 W

Powercap Impact on Performance-Energy Tradeoff: A100



- same peaks in performance per watt in the frequency benchmark

Powercap Impact on Performance-Energy Tradeoff: A40



Same efficiency curve shift to the left can be observed for the powercap benchmarks

Conclusion and future plans



Conclusion

Overall

Energy-to-solution is dominated by **configuration and workload characteristics**

- Optimal settings differ between CPU and GPU
- Frequency plays a secondary role

CPU

- No major scaling bottleneck observed; **memory bandwidth did not saturate** inside one ccNuma domain
- Frequency had a comparatively small impact

GPU

- AMBER GPU runs behaved as **memory/throughput-limited** based on number of Molecules/Atoms in Input file
- GPU power increases **nonlinearly** with frequency
- **Power cap plays a bigger role than frequency**

Future work

Better understand the performance and energy characteristics by extending:

- CPU scaling to multi-node systems to capture bandwidth and communication limits.
- GPU-based runs on NVIDIA L40, H100, and GH200 for architectural comparison.
- Tests on additional applications such as Firestarter which runs at near-peak power consumption: <https://tu-dresden.de/zih/forschung/projekte/firestarter>

Thank you. Any Questions?



References

- <https://doc.nhr.fau.de/clusters/alex/>
- <https://doc.nhr.fau.de/clusters/fritz/>
- A. Afzal, G. Hager, and G. Wellein, “Analytic roofline modeling and energy analysis of the LULESH proxy application on multi-core clusters,”
- A. Afzal, G. Hager, and G. Wellein, “SPEChpc 2021 Benchmarks on Ice Lake and Sapphire Rapids Infiniband Clusters: A Performance and Energy Case Study,”
- A. Afzal, A. Kahler, G. Hager, and G. Wellein, “GROMACS Unplugged: How Power Capping and Frequency Shapes Performance on GPUs,”